

GEOCHANG, Korea, Republic of

WMO: 471570

Lat: 35.6674N

Lon: 127.9100E

Elev: 226

StdP: 98.64

Time Zone: 9.00 (E09)

Period: 09-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-10.9	-9.3	-21.3	0.6	-3.5	-18.9	0.7	-1.9	6.4	1.2	5.8	1.7	1.2	250	0.383

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	8.2	32.8	24.4	31.5	23.8	30.0	22.9	25.8	30.3	25.2	29.4	24.7	28.3	2.2	140

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
24.7	20.2	27.4	24.1	19.6	26.9	23.6	18.9	26.5	81.2	30.5	78.7	29.6	76.3	28.5	27.8

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 6.0	5.2	4.5	DB	-13.8	34.5	1.9	0.9	-15.2	35.1	-16.3	35.6	-17.4	36.1	-18.8	36.7	(4)
(5)			WB	-14.4	26.8	1.9	0.8	-15.8	27.4	-16.9	27.8	-17.9	28.2	-19.3	28.8	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	12.5	-1.4	1.4	6.2	11.7	17.5	21.7	25.0	25.2	20.2	13.9	7.4	0.8	(6)	
	DBStd	9.68	3.45	3.67	3.63	3.35	2.59	2.03	2.34	2.55	2.70	3.15	3.83	3.50	(7)	
	HDD10.0	1132	352	241	128	21	0	0	0	0	0	6	98	287	(8)	
	HDD18.3	2734	611	473	376	199	47	2	0	0	11	142	329	545	(9)	
	CDD10.0	2053	0	2	10	73	232	350	464	472	306	127	18	0	(10)	
	CDD18.3	614	0	0	0	1	21	102	206	213	67	5	0	0	(11)	
	CDH23.3	4843	0	0	1	38	342	719	1626	1733	371	14	0	0	(12)	
	CDH26.7	1506	0	0	0	2	67	174	563	639	59	1	0	0	(13)	
(14) Wind	WSAvg	1.5	1.6	1.6	1.9	1.9	1.8	1.5	1.3	1.3	1.1	1.2	1.4	1.6	(14)	
(15) Precipitation	PrecAvg	1264	27	40	56	96	86	171	292	246	136	52	40	21	(15)	
	PrecMax	1958	137	107	101	195	164	382	560	464	376	144	92	48	(16)	
	PrecMin	688	0	0	4	30	4	3	79	46	5	1	4	0	(17)	
	PrecStd	330	31	33	23	53	44	93	141	122	95	41	28	15	(18)	
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	10.6	15.7	21.7	26.5	31.0	32.0	33.8	34.5	30.2	25.1	20.5	13.1	(19)	
		MCWB	4.6	11.2	11.7	13.7	16.9	20.0	25.0	25.1	23.0	17.9	14.9	9.3	(20)	
	2%	DB	8.2	13.0	18.7	24.2	28.4	30.0	32.7	33.2	28.3	22.9	17.7	10.4	(21)	
		MCWB	3.2	8.1	9.5	13.5	16.2	19.7	24.9	25.1	21.5	16.2	12.7	5.7	(22)	
	5%	DB	6.2	10.6	16.2	22.0	26.7	28.6	31.3	31.9	26.8	21.5	15.9	8.7	(23)	
		MCWB	2.0	4.8	8.3	11.8	15.6	19.4	24.2	24.6	20.4	15.5	11.6	5.0	(24)	
	10%	DB	4.7	8.3	13.8	19.6	25.3	27.4	29.8	30.3	25.6	20.1	14.1	6.7	(25)	
		MCWB	0.8	3.3	7.8	10.7	15.5	19.4	23.6	24.2	19.8	15.1	10.3	3.6	(26)	
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	6.7	13.4	13.5	16.9	21.6	23.7	26.6	26.6	24.6	20.7	16.9	11.0	(27)	
		MCDB	9.4	14.5	17.2	22.4	23.7	26.9	30.5	31.7	28.2	22.6	19.6	13.2	(28)	
	2%	WB	4.1	8.7	11.9	14.8	19.2	22.7	25.9	25.9	23.5	19.1	14.6	7.2	(29)	
		MCDB	6.8	11.2	15.3	20.8	24.3	26.2	30.1	30.8	26.1	21.3	16.3	9.2	(30)	
	5%	WB	2.8	5.7	10.3	13.6	17.8	21.6	25.3	25.4	22.6	17.6	12.7	5.5	(31)	
		MCDB	5.4	8.9	14.6	18.3	23.3	25.2	29.2	29.9	25.0	19.4	14.9	7.7	(32)	
	10%	WB	1.3	4.0	8.4	12.4	16.7	20.8	24.7	24.9	21.4	15.8	11.1	4.0	(33)	
		MCDB	4.2	7.8	13.0	17.3	22.0	24.6	28.3	28.9	23.9	18.7	13.2	6.8	(34)	
(35) Mean Daily Temperature Range	5% DB	MDBR	11.1	11.3	12.6	13.1	13.5	10.1	7.9	8.2	9.7	12.3	11.4	10.4	(35)	
		MCDBR	13.3	14.8	17.2	18.2	17.4	13.3	10.4	11.1	11.3	13.8	13.9	13.1	(36)	
	5% WB	MCWBR	8.6	9.2	8.9	8.0	6.4	4.4	3.4	3.6	4.6	6.7	8.2	8.8	(37)	
		MCWBR	11.6	12.4	13.8	13.0	12.4	8.4	8.8	9.1	7.9	8.5	10.1	11.4	(38)	
(40) Clear-Sky Solar Irradiance	taub		0.434	0.485	0.573	0.603	0.614	0.676	0.625	0.579	0.504	0.476	0.453	0.429	(40)	
		taud	1.936	1.858	1.693	1.677	1.691	1.630	1.799	1.899	2.042	1.993	1.973	1.967	(41)	
	Ebn at Noon		716	728	700	709	710	665	697	720	755	730	692	689	(42)	
		Edh at Noon	145	176	226	241	241	256	215	191	158	151	139	133	(43)	
(44) All-Sky Solar Radiation	RadAvg		2.66	3.37	4.37	4.99	5.54	4.93	4.31	4.39	3.96	3.63	2.69	2.37	(44)	
	RadStd		0.18	0.29	0.41	0.39	0.53	0.44	0.60	0.59	0.43	0.33	0.31	0.15	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (45 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+68	N/A

Nomenclature: See separate page